

Seungmin Baik

(201) 359-4298 | sb1972@scarletmail.rutgers.edu | New Brunswick, NJ 08901 | www.linkedin.com/in/seungmin-baik/

EDUCATION

Rutgers, The State University of New Jersey

Bachelor of Science in Electrical and Computer Engineering

GPA : 3.845/4.00

New Brunswick, NJ

Expected Graduation: May 2026

Relevant Coursework: Digital Logic Designs, Electronic Devices, Computer Architectures, Data Structure, Digital Electronics, Embedded System, Analog Electronics, Sustainable Energy, Intro to Deep Learning

PROFESSIONAL EXPERIENCE & PROJECTS

Rutgers University Research - BIC (Beyond Intelligent Computing) Lab

New Brunswick, NJ

Hardware Chip Research Assistant

May 2025 – Present

- Conducting experiments on layer-wise precision scaling in Convolution Neural Networks using PyTorch, achieving a 49% reduction in total bit usage with only a 2% accuracy degradation compared to a uniform-bit-width baseline
- Reviewed 5+ pre-submission manuscripts on AI-accelerator architectures as a secondary reviewer, providing constructive and detailed feedback to strengthen alignment, coherence, and overall research quality and clarity

Rutgers University Research - Aresty Program

New Brunswick, NJ

AI Training & Data Annotation Research Assistant

October 2025 – Present

- Developing a Python automation tool that detects traffic-light signal changes and automatically captures and stores transition frames to streamline dataset collection and reduce manual extraction time for AI model training
- Annotated 750+ rail-footage images with object-level bounding labels such as vehicles, pedestrians, trains, and rail gates to build a high-quality dataset supporting AI-based trespassing detection and railway safety applications

Capstone Design Project - SoleSignal

New Brunswick, NJ

Signal Processing & Data Analysis Developer

September 2025 – Present

- Implemented an Arduino-based real-time signal processing pipeline that analyzes streaming FSR sensor data to distinguish walking and running gait patterns
- Applied Exponential Moving Average (EMA) filtering to reduce sensor noise and stabilize real-time pressure signals during dynamic motion

Analog Electronics Project - Two Stage Op-Amp Design

New Brunswick, NJ

Theoretical Design Lead

September 2025 – December 2025

- Performed first-principles hand calculations based on MOSFET equations to derive transistor sizing (W/L), bias currents, and small-signal parameters (gm) for a two-stage CMOS op-amp meeting design specifications
- Confirmed theoretical predictions through transistor-level simulations in Cadence Virtuoso by validating DC operating points and small-signal response, and refining design parameters based on simulation results

Embedded System Project - Smart Mini Greenhouse

New Brunswick, NJ

FPGA Developer

January 2025 – May 2025

- Integrated 4+ Pmods with FPGA to build an environmental control system that preserve temperature and humidity
- Designed and simulated threshold-based control logic in VHDL, verified with custom testbenches using Vivado

PACT - Human Trafficking Awareness Organization

New Brunswick, NJ

Webpage Interface Design Externship Team Lead

September 2024 – December 2024

- Utilized WordPress to design anti-human trafficking website's structure and create outline for future improvements
- Communicated with advisors and team members to monitor progress and ensure project completion by the deadlines
- Prepared weekly progress updates as part of an externship course, summarizing project milestones and deliverables

LEADERSHIP

Korean Army

South Korea Yeon-Cheon

5th Infantry Division Reconnaissance Squad Leader

April 2021 – October 2022

- Managed squad members' health, logistics, and equipment to ensure operational readiness under mission conditions
- Secured and maintained top-secret and classified unit documents with quarterly updates and protocol compliance
- Developed 100+ operational plans for DMZ missions and allocated required personnel and equipment accordingly

SKILLS & AWARDS

Languages & Technical Skills : English | Korean(fluent) | Python | Java | Javascript | VHDL | FPGA

Proficiencies: Leadership | Teamwork | Communication | Responsibility